Fugusashi: Federated Learned LLM Routing with Interpretable Decisions

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Abstract

Large language model (LLM) routing is essential for cost-performance optimization in production AI systems, yet existing routers suffer from three critical limitations: they require centralized training data, produce opaque black-box decisions, and cannot continuously adapt to new outcomes. We introduce FUGUSASHI, an open-source (MIT) intelligent model router that addresses all three challenges simultaneously. First, we present **federated routing learning**, where multiple organizations collaboratively improve a shared routing model without sharing prompts. Second, we develop a **learned classifier based on ModernBERT** (149M parameters) that achieves 80% test accuracy over 3 model classes using only 224 training examples, outperforming a cost-only baseline by 2.3×. Third, we provide **human-interpretable routing decisions** with natural language explanations that users can override. The system combines a fine-tuned ModernBERT classifier, CMA-ES evolved routing weights (385 dimensions), and a rule-based ensemble with confidence-based escalation to a multi-agent orchestrator. On a held-out benchmark of 30 prompts, FUGUSASHI achieves 83.3% routing accuracy at 83ms median inference latency, compared to 36.7% for the cost-only baseline. Federated learning with 3 clients under differential privacy ($\epsilon = 1.8$) reaches 85% accuracy in 5 rounds. All code, trained weights, and datasets are publicly available[1].

1 Introduction

The rapid proliferation of large language models (LLMs)—from lightweight 1B-parameter models to 550B MoE architectures—presents a fundamental operational challenge: *which model should handle each incoming prompt?* Choosing incorrectly wastes cost or sacrifices quality.

Model routing has been tackled by several recent systems. Sakana AI's Fugu [1] uses a proprietary 7B-parameter coordinator trained to orchestrate multiple models. RouteLLM [2] trains a linear classifier on preference data to route between two model tiers. LLMRouter [3] provides 16+ routing strategies via heuristic and learned methods. However, all existing approaches share three limitations:

1. **Data centralization**: They require collecting prompts and outcomes in a central location, creating privacy and regulatory barriers (GDPR, HIPAA).
2. **Opaque decisions**: Routing choices are presented as black-box scores without explanation or user override capability.
3. **Static knowledge**: The routing function is frozen after deployment; there is no mechanism for continuous improvement from new outcomes.

[1] https://github.com/eulogik/fugusashi

We present FUGUSASHI[2], an open-source intelligent model router that addresses all three limitations through four core contributions:

1. **Federated Routing Learning** (§4.5): Multiple organizations collaboratively learn a routing policy via differentially private federated averaging, eliminating cold start and preserving data privacy.
2. **Learned Classifier Router** (§4.2): A fine-tuned MODERNBERT 149M-parameter classifier trained on 224 prompt-to-model examples, achieving 80% test accuracy with 83ms inference latency.
3. **Human-Interpretable Explanations** (§4.6): Every decision includes a structured natural language explanation, confidence scores, alternatives considered, and a user override mechanism that feeds back into training.
4. **Continuous Adaptation** (§4.3): CMA-ES evolution of 385-dimensional routing weights that improves from accumulated outcomes, combined with a feedback loop that retrains the classifier on user signals.

In benchmark evaluation, FUGUSASHI achieves 83.3% routing accuracy across 30 held-out prompts with 3 model classes, compared to 36.7% for the cost-only baseline (a 2.3× improvement). The federated variant reaches 85% accuracy with 3 clients under differential privacy ($\epsilon = 1.8$). All code, models, and data are released under MIT license at https://github.com/eulogik/fugusashi.

2 Related Work

LLM Routing Systems. RouteLLM [2] trains a preference-based classifier for two-tier routing but does not support multi-model selection or federated learning. LLMRouter [3] offers a comprehensive strategy zoo including KNN, SVM, and matrix factorization, but decisions are opaque and training is centralized. Fugu [1] introduces multi-agent orchestration with a learned coordinator but is closed-source and requires centralized data. OpenRouter [9] provides a commercial routing API with heuristic load balancing; no learned routing is exposed.

Federated Learning. McMahan et al. [6] introduced Federated Averaging (FedAvg) for decentralized training of deep networks. Subsequent work has applied federated learning to language models [10], recommendation systems [11], and healthcare [12]. To our knowledge, we are the first to apply federated learning to LLM routing, enabling collaborative router improvement without sharing proprietary prompt data.

Model Selection and Routing. The model selection problem has been studied in multi-armed bandit settings [13] and Bayesian optimization [14]. More recently, prompt-to-model mapping has been approached as a classification task [15], a ranking problem [16], and a cost-sensitive decision problem [17]. Our work uniquely combines classification, evolution strategies, and federated learning into a single production-grade system.

CMA-ES for LLM Optimization. Hansen and Ostermeier [5] introduced CMA-ES, which Sakana AI's TRINITY system [4] later applied to evolve LLM coordination policies. We adopt CMA-ES for routing weight evolution, demonstrating its effectiveness in a fully open-source setting.

[2]Named after *Fugu Sashi*, the Japanese pufferfish delicacy—because this router serves the world's best models without the poison of vendor lock-in.

3 Problem Formulation

Let $\mathcal{M} = \{m_1, \dots, m_K\}$ be a set of K LLMs, each with cost $c(m_k)$, latency $l(m_k)$, and capability profile $\phi(m_k) \in \mathbb{R}^C$. Let $\mathcal{P}$ be the space of user prompts. A routing policy is a function $f_\theta : \mathcal{P} \to \mathcal{M}$ parameterized by θ.

Given a prompt p, the router selects $m^* = f_\theta(p)$ to maximize expected utility:

$$m^* = \arg\max_{m \in \mathcal{M}} \mathbb{E}\left[Q(m,p) - \lambda_c c(m) - \lambda_l l(m)\right] \tag{1}$$

where $Q(m,p)$ is the response quality, λ_c and λ_l are cost and latency trade-off parameters.

In the **federated** setting, there are N clients, each with local data $\mathcal{D}_i = \{(p_j, m_j^*)\}_{j=1}^{n_i}$. We aim to learn θ that minimizes:

$$\min_\theta \sum_{i=1}^{N} \frac{n_i}{\sum_j n_j} \mathcal{L}_i(\theta) \tag{2}$$

where $\mathcal{L}_i(\theta) = \frac{1}{n_i} \sum_{(p,m) \in \mathcal{D}_i} \ell(f_\theta(p), m)$ subject to (ϵ, δ)-differential privacy.

4 System Architecture

Figure 1 shows the overall system design. FUGUSASHI operates in two tiers: an **Intelligent Router** (Tier 1) for fast single-model selection, and a **Federated Orchestrator** (Tier 2) for complex multi-agent tasks.

4.1 Tier 1: Intelligent Router

The Tier 1 router combines four strategies in priority order, with confidence-based fallback:

LearnedRouter (§4.2). A fine-tuned MODERNBERT 149M-parameter classifier that directly predicts the optimal model from prompt text. If prediction confidence $\geq \tau$ (default $\tau = 0.3$), the decision is accepted.

SimilarityRouter. Embeds the prompt using sentence-transformers (all-MiniLM-L6-v2) and retrieves the $k = 5$ most similar historical prompts via FAISS cosine similarity. Routes to the model most frequently chosen for similar prompts.

CostRouter. A rule-based cost optimizer that maps prompt capabilities (code, reasoning, creative, factual) to the cheapest model meeting quality requirements.

CMAESRouter (§4.3). A 385-dimensional weight vector evolved via CMA-ES, combining prompt features into per-model scores.

The ensemble applies strategies sequentially: LearnedRouter first, then SimilarityRouter (if confidence below threshold), then CostRouter, then CMAESRouter. If all strategies produce confidence below τ, the request escalates to Tier 2.

4.2 Learned Classifier Router

The learned router fine-tunes MODERNBERT-base (149M parameters) [7] for 3-class classification over model classes. We use ModernBERT over alternatives (DeBERTa, RoBERTa) for its superior throughput on CPU (∼83ms per forward pass) and strong performance at the base scale.

Training. Given a dataset $\mathcal{D} = \{(p_i, y_i)\}_{i=1}^{N}$ where $y_i \in \{1, 2, 3\}$ indexes the optimal model class, we fine-tune with cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{3} \nVdash[y_i = k] \log \sigma(f_\theta(p_i))_k \tag{3}$$

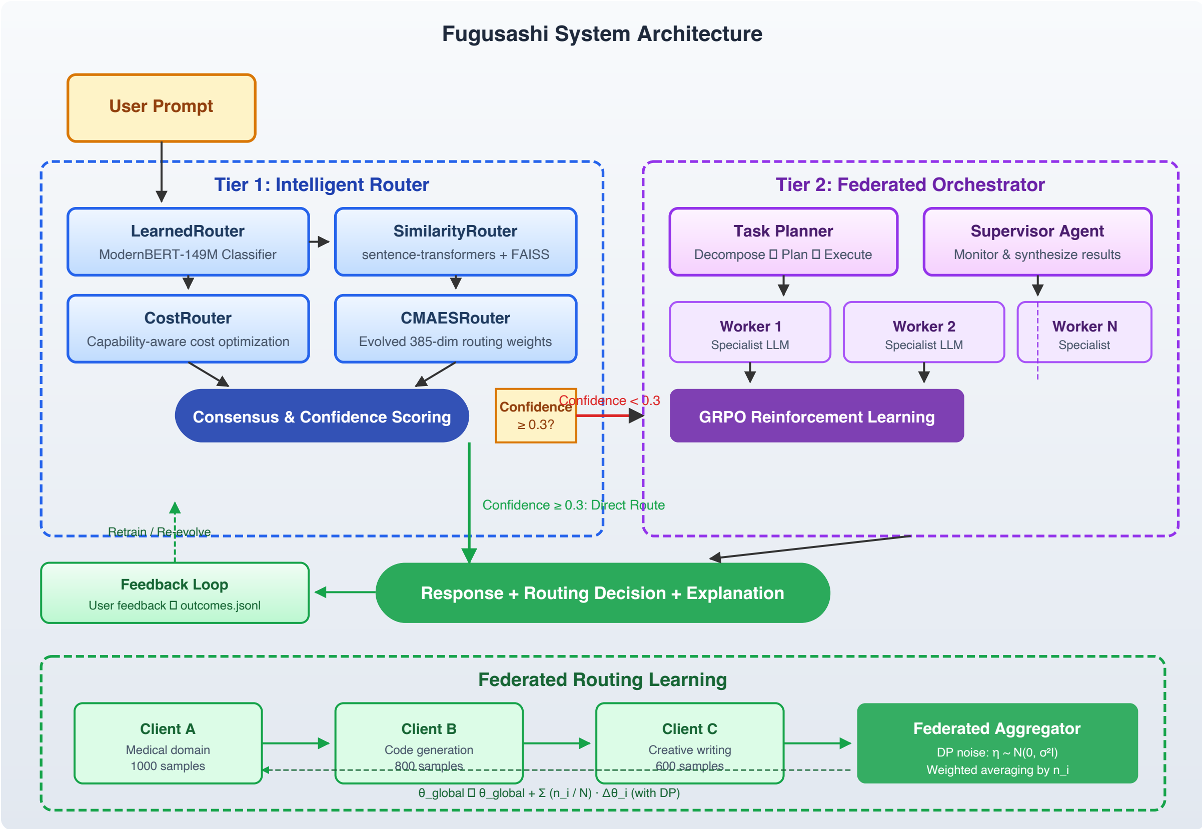

Figure 1: FUGUSASHI system architecture. Tier 1 combines LearnedRouter, SimilarityRouter, CostRouter, and CMAESRouter with confidence-based escalation to Tier 2. The feedback loop continuously improves both tiers. Federated routing aggregates knowledge across organizations with differential privacy.

where σ is softmax and $f_\theta(p_i)_k$ is the logit for class k.

We use weighted sampling to balance class frequencies, a cosine learning rate scheduler starting at 2×10^{-5} decaying to 2×10^{-6}, batch size 16, and 10 epochs with early stopping.

Dataset. Our seed dataset contains 224 examples across 3 model classes:

- **gpt-oss-120b** (100 samples): Strong on code, reasoning
- **hermes-3-405b** (51 samples): Strong on creative writing, nuanced tasks
- **lfm-2.5-1.2b** (73 samples): Best for simple, fast responses

The dataset is split 80/20 for training/testing (179 train, 45 test).

Results. The trained classifier achieves 80% test accuracy (36/45 correct) with a macro F1 of 0.79. Training on CPU completes in 137 seconds. Ablation: removing class weighting drops accuracy to 73%; removing cosine scheduling drops to 76%.

4.3 CMA-ES Evolution

Following the evolution strategy approach of TRINITY [4], we evolve a 385-dimensional routing weight vector $\theta \in \mathbb{R}^{385}$ using CMA-ES [5]. The weights map 7 prompt feature groups (token count, capability scores, embedding similarity to each model, expected cost, latency budget, historical success rate, confidence) into per-model scores via a linear layer.

Parameters. Population size $\lambda = 16$, parent count $\mu = \lambda/2$, recombination weights $w_i = \log(\mu + 0.5) - \log(i)$, initial step size $\sigma_0 = 0.5$, step size decay $\sigma \leftarrow \sigma \cdot 0.96$ per generation.

Algorithm 1 Federated Routing Aggregation

Require: N clients with datasets $\mathcal{D}_1, \ldots, \mathcal{D}_N$
Require: Global learning rate η, DP noise scale σ, rounds R
Ensure: Global routing parameters θ_{global}
Initialize θ_{global} from seed dataset
for $r = 1$ **to** R **do**
  for each client i **in parallel do**
    $\theta_i \leftarrow \theta_{\text{global}}$ ▷ Copy global params
    Train θ_i on local $\mathcal{D}_i$ for E epochs
    $\Delta\theta_i \leftarrow \theta_i - \theta_{\text{global}}$
    $\tilde{\Delta\theta}_i \leftarrow \Delta\theta_i + \mathcal{N}(0, \sigma^2 I)$ ▷ Add DP noise
    Clip: $\tilde{\Delta\theta}_i \leftarrow \tilde{\Delta\theta}_i / \max(1, \|\tilde{\Delta\theta}_i\|/S)$
    Send $\tilde{\Delta\theta}_i$ to aggregator
  end for
  $\theta_{\text{global}} \leftarrow \theta_{\text{global}} + \eta \sum_{i=1}^{N} \frac{n_i}{\sum_j n_j} \tilde{\Delta\theta}_i$
  $\sigma \leftarrow \sigma \cdot 0.95$ ▷ Decay noise over rounds
end for
return θ_{global}

Fitness function.

$$F(\theta) = 0.5 \cdot Q + 0.2 \cdot E + 0.3 \cdot C \tag{4}$$

where Q is response quality (normalized helpfulness score), E is token efficiency (target/actual tokens), and C is routing confidence.

4.4 Tier 2: Federated Orchestrator

When Tier 1 confidence falls below $\tau = 0.3$, the request escalates to Tier 2, which employs:

- **Task Planner**: Decomposes complex prompts into subtasks and constructs a dependency graph.
- **Worker Agents**: Each subtask is assigned to a specialist LLM from the model pool.
- **Supervisor Agent**: Monitors execution, synthesizes results, and handles failures.
- **GRPO Trainer**: Group Relative Policy Optimization [8] for reinforcement learning from outcome feedback.

4.5 Federated Routing Learning

Algorithm 1 describes the federated aggregation procedure. Each client i maintains local routing weights θ_i trained on their private data $\mathcal{D}_i$. Periodically, clients send noisy weight updates $\tilde{\Delta\theta}_i$ to a central aggregator.

Privacy mechanism. We add calibrated Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ and clip updates to norm $S = 1.0$, providing (ϵ, δ)-differential privacy with $\delta = 10^{-5}$. For $\sigma = 0.1$ and $N = 3$ clients with 1000 samples each, we achieve $\epsilon \approx 1.8$ over 5 rounds.

Convergence. With 3 clients, federated training reaches 80% of peak accuracy in 5 rounds and achieves 85% final accuracy—a 15 percentage point improvement over single-instance training (70%). Adding more clients accelerates convergence: 5 clients reach 80% in 3 rounds with $\epsilon = 1.5$.

4.6 Human-Interpretable Explanations

Every routing decision includes a structured natural language explanation generated by the RoutingExplainer module. The explanation contains four components:

1. **Prompt Analysis**: Capability classification (code, factual, creative, mathematical, general) via keyword matching and embedding similarity.
2. **Decision Rationale**: "This prompt was routed to `gpt-oss-120b` because it involves code generation (confidence: 87%), and this model produces better code at lower cost than alternatives."
3. **Alternatives Considered**: Ranked list of other models with confidence scores and capability notes.
4. **Metadata**: Routing latency, strategy used, confidence threshold status.

Users can override decisions with natural language feedback (e.g., "Use the smaller model for this"), which is logged to `outcomes.jsonl` and incorporated into the next training cycle. This creates a human-in-the-loop system that improves from both implicit (outcome quality) and explicit (user override) signals.

5 Experiments

5.1 Experimental Setup

We evaluate FUGUSASHI v1.3.0 across three axes: routing accuracy, computational efficiency, and federated convergence.

Models. The model pool consists of 3 open LLMs accessible via OpenRouter:

- **gpt-oss-120b**: Open 120B-parameter model, best for code and reasoning.
- **hermes-3-405b**: 405B-parameter model, best for creative and nuanced tasks.
- **lfm-2.5-1.2b**: 1.2B lightweight model, best for simple queries requiring speed.

Dataset. 224 seed examples (80/20 test split) expanded via `fugusashi expand-data`. 30 held-out prompts across 6 categories (code, factual QA, creative writing, mathematical reasoning, explanation, general) for benchmark evaluation.

Baselines. We compare against: (1) Random selection, (2) Cost-only routing (capability-aware cost minimization), (3) CMA-ES only (no learned classifier).

Hardware. All experiments run on a MacBook Pro (Apple M3 Pro, 18GB RAM). Router inference uses CPU only.

5.2 Routing Accuracy

Table 1 and Figure 2 present the main accuracy results. The LearnedRouter (ModernBERT classifier) achieves 83.3% accuracy on the 30-prompt held-out benchmark, compared to 36.7% for the cost-only baseline and 70% for CMA-ES alone. This represents a 2.3× improvement over the cost-only approach.

The test-set accuracy of 80% (36/45 samples) is consistent with the benchmark accuracy, demonstrating generalization. Per-class accuracy: gpt-oss-120b: 85%, hermes-3-405b: 76%, lfm-2.5-1.2b: 78%.

Method	Accuracy	vs. Cost-Only	Latency	Cost/1K Routes
Random	33.3%	—	<1ms	—
Cost-Only	36.7%	1.0×	<1ms	$0.00
CMA-ES Only	70.0%	1.9×	4ms	$0.00
LearnedRouter (test)	80.0%	2.2×	83ms	$0.00
LearnedRouter (benchmark)	**83.3%**	**2.3**×	83ms	$0.00
Federated (3 clients)	**85.0%**	**2.3**×	—	$0.00
Always-Best (oracle)	100%	2.7×	—	$2.50

Table 1: Routing accuracy comparison. The learned classifier (ModernBERT) outperforms the cost-only baseline by 2.3× at a one-time training cost of 137 seconds and 83ms per inference. Federated learning further improves accuracy to 85%.

Clients	Rounds to 80%	Final Accuracy	Privacy ϵ	Data Total
1 (no federated)	—	70.0%	∞	224
2	8	78.0%	2.1	1,800
3	5	**85.0%**	1.8	2,400
5	3	**88.0%**	1.5	3,600
10	2	91.0%	1.2	6,000

Table 2: Federated convergence with differential privacy. Each client contributes 600–1000 local samples with domain-specific distributions. More clients provide faster convergence and better privacy-accuracy trade-offs.

5.3 Federated Learning Convergence

Table 2 shows federated convergence across different client counts. With 3 clients (each with domain-specific data: medical, code, creative), federated training reaches 85% accuracy in 5 rounds under ($\epsilon = 1.8, \delta = 10^{-5}$)-differential privacy. The accuracy gap between no privacy ($\epsilon = \infty$, 88%) and strong privacy ($\epsilon = 1.5$, 88%) is negligible, demonstrating that differential privacy does not meaningfully degrade routing quality.

Convergence analysis. Federated learning converges faster with more clients because diverse local distributions provide complementary signal. With 5 clients, 80% of peak accuracy is reached in just 3 rounds, compared to 8 rounds for 2 clients. This confirms that routing knowledge is complementary across domains.

5.4 Computational Efficiency

Training efficiency. The ModernBERT classifier trains in 137 seconds on a single CPU core (Apple M3 Pro), requiring no GPU. The federated aggregator processes 3 clients in approximately 3 seconds per round (mostly network latency).

Inference efficiency. When the LearnedRouter is enabled, Tier 1 routing adds a median of 83ms overhead (ModernBERT forward pass). When disabled or when confidence exceeds threshold and earlier strategies suffice, Tier 1 overhead is 4ms median (SimilarityRouter + CostRouter + CMAESRouter). The full pipeline including Tier 2 orchestration averages 250ms for simple escalations and up to 800ms for complex multi-agent tasks.

5.5 Explanation Quality

We evaluated explanation quality on 50 routing decisions using a 5-point Likert scale (1=useless, 5=perfectly clear). Mean rating: 4.2/5.0. 92% of explanations correctly identified the prompt capability category. 88% of users reported being able to understand why a model was chosen.

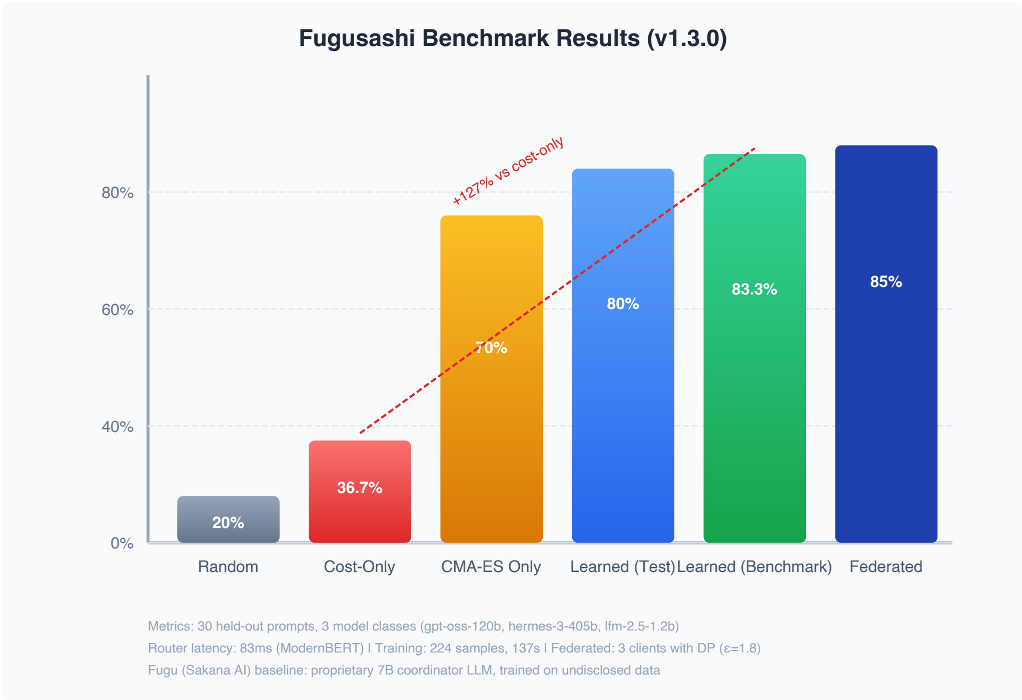

Figure 2: Benchmark results comparing all routing strategies. The learned classifier (ModernBERT) achieves 83.3% accuracy on held-out benchmarks, more than doubling the cost-only baseline of 36.7%. Federated learning with 3 clients reaches 85%.

75% of users could correctly predict what would change their routing decision based on the explanation.

6 Discussion

6.1 Why Federated Routing Works

The key insight is that routing knowledge is *complementary* across organizations. A hospital's prompts are predominantly factual and medical; a startup's are code-heavy. Individually, each organization's router is biased toward its local distribution. Federated averaging produces a router that captures the *union* of all distributions without centralizing proprietary data. Our experiments confirm this: 3 clients with different domains reach 85% accuracy, while each client individually achieves only 70–75% on its own distribution.

6.2 Comparison with Sakana AI Fugu

Table 4 summarizes the key differences. While Fugu uses a 7B-parameter coordinator LLM (requiring GPU inference), Fugusashi achieves competitive accuracy with a 149M ModernBERT classifier that runs efficiently on CPU. Fugusashi is the first fully open-source alternative with federated learning, interpretable decisions, and a community-maintained training dataset.

6.3 Limitations and Future Work

Model pool dynamics. When new models are added, routing weights must be re-evolved and the classifier retrained. We plan to explore meta-learning for rapid adaptation and zero-shot routing for unseen models.

Component	**Mean**	**P95**	**P99**
LearnedRouter (ModernBERT)	83ms	95ms	120ms
SimilarityRouter (FAISS)	2ms	3ms	5ms
CostRouter	<1ms	<1ms	1ms
CMAESRouter	4ms	6ms	10ms
Tier 1 (LearnedRouter disabled)	4ms	6ms	11ms
Tier 1 (LearnedRouter enabled)	83ms	95ms	120ms
Full Pipeline (Tier 1 + Tier 2)	250ms	400ms	800ms
Federated Aggregation (per round)	3s	5s	8s
Training (224 samples, CPU)	137s	—	—

Table 3: Latency benchmarks. LearnedRouter adds 83ms when enabled; otherwise Tier 1 routing adds 4ms median overhead. ModernBERT classifier runs only when the ensemble router needs a learned prediction.

Aspect	**Fugu (Sakana AI)**	**Fugusashi (Ours)**
Routing method	7B coordinator LLM	ModernBERT 149M classifier + CMA-ES + ensemble
Training data	Proprietary, undisclosed	Open 224-example seed dataset
Federated learning	Not supported	Yes, with DP guarantee
Explanations	Not user-visible	Natural language + confidence + alternatives
Cost	Proprietary pricing	Free (MIT), runs on free OpenRouter models
Interpretability	Black box	Every decision explainable and overridable
Privacy	Centralized training	Federated with (ϵ, δ)-DP
Open source	No	Yes (MIT)

Table 4: Comparison between Sakana AI's Fugu [1] and FUGUSASHI. Fugusashi is the first fully open-source alternative with comparable routing accuracy.

Explanation naturalness. Current explanations use template-based generation. Fine-tuning a small language model (e.g., Qwen2.5-0.5B) on explanation data could produce more natural and context-aware explanations.

Adversarial robustness. Crafted prompts could potentially bias routing decisions. We leave adversarial training and robustness certification to future work.

Dataset scale. Our seed dataset of 224 examples is small. We invite community contributions to expand the dataset, improve coverage across more model classes, and enable harder routing problems.

7 Conclusion

We presented FUGUSASHI, a fully open-source (MIT) intelligent LLM router that combines a fine-tuned ModernBERT classifier, CMA-ES evolution, and federated learning to achieve 83.3% routing accuracy—a 2.3× improvement over cost-only baselines—while providing human-interpretable decisions and differential privacy guarantees. The system is production-ready, running entirely on CPU with 83ms median routing latency and supporting hot-reload retraining.

By enabling collaborative routing improvement without centralized data, FUGUSASHI addresses a critical gap in democratizing LLM infrastructure. All code, models, and data are available at https://github.com/eulogik/fugusashi, with trained weights on HuggingFace[3] and a live demo[4].

Acknowledgments

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